Obligatory Exercises 2

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Exercise O2.1

$F_1 : ((p \to q) \wedge (q \to r)) \to (p \to r)$

To prove the validity of F_1 in the resolution calculus we can prove that it is not the opposite of that instead, namely: that the negated F_1 is unsatisfiable instead. But first, we need to transform the formula into clausal form.

$$\begin{aligned}
&\neg(((p \to q) \wedge (q \to r)) \to (p \to r)) & \\
&\equiv ((p \to q) \wedge (q \to r)) \wedge \neg(p \to r) & Using\ \neg(A \to B) \equiv A \wedge \neg B \\
&\equiv ((p \to q) \wedge (q \to r)) \wedge (p \wedge \neg r) & Using\ \neg(A \to B) \equiv A \wedge \neg B \\
&\equiv ((\neg p \vee q) \wedge (\neg q \vee r)) \wedge (p \wedge \neg r) & Using\ A \to B \equiv \neg A \vee B
\end{aligned}$$

Thus the resulting set of clauses becomes $\{\{\neg p, q\}, \{\neg q, r\}, \{p\}, \{\neg r\}\}$

1. $\neg p, q$
2. $\neg q, r$
3. p
4. $\neg r$
5. q − Resolvent of 1. and 3.
6. r − Resolvent of 2. and 5.
7. $\Box$ − Resolvent of 4. and 6.

And just like that, by proving that the negated form of F_1 is unsatisfiable we can conclude that F_1 is valid.

$F_2 : \forall x\ \exists y\ (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z\ q(z)$

To prove the validity of F_2 we do the exact same as we did with F_1. So, we convert the negated F_2 into clausal form first.

$$
\begin{array}{ll}
\neg(\forall x\ \exists y\ (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z\ q(z)) & \\
\equiv \forall x\ \exists y\ (p(x) \wedge (p(y) \rightarrow q(x))) \wedge \neg\forall z\ q(z) & Using\ \neg(A \rightarrow B) \equiv A \wedge \neg B \\
\equiv \forall x\ \exists y\ (p(x) \wedge (\neg p(y) \vee q(x))) \wedge \neg\forall z\ q(z) & Using\ A \rightarrow B \equiv \neg A \vee B \\
\equiv \forall x\ \exists y\ (p(x) \wedge (\neg p(y) \vee q(x))) \wedge \exists z\ \neg q(z) & Using\ \neg\forall x\ A \equiv \exists x\ \neg A \\
\equiv \forall x\ \exists y\ \exists z\ ((p(x) \wedge (\neg p(y) \vee q(x))) \wedge \neg q(z)) & Using\ \neg\forall x\ A \equiv \exists x\ \neg A \\
\equiv \forall x\ ((p(x) \wedge (\neg p(f(x)) \vee q(x))) \wedge \neg q(g(x))) & Using\ Skolemisation
\end{array}
$$

Thus the resulting set of clauses becomes $\{\{p(x)\}, \{\neg p(f(x)), q(x)\}, \{\neg q(g(x))\}$

1. $p(x)$
2. $\neg p(f(x)), q(x)$
3. $\neg q(g(x))$
4. $\neg p(f(x))$ $-$ Resolvent of 2. and 3. with $\sigma = [x \backslash g(x)]$
5. $\square$ $-$ Resolvent of 1. and 4. with $\sigma = [x \backslash f(x)]$

This proves the unsatisfiability of the negated form of F_2 which as a result proves the validity of F_2.

Exercise O2.2

In order for the programme to accept this new operator $\uparrow$ and be able to transform it into the clausal form we can transform all its instances into its equivalent, namely: $A \uparrow B$ into $\neg(A \wedge B)$, which the programme already knows how to handle. From there we can use the NNF to transform it further into $\neg A \vee \neg B$ which transforms directly into $\{\neg A, \neg B\}$, the clausal form. Although the program may not be able to transform into clausal right away since mulitple $\uparrow$ operators may be present.